Topology-Based Coarse-Graining for Multifractal Spectrum Analysis in Metal-Organic Frameworks

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Metal-organic frameworks (MOFs) are porous crystals whose behavior depends strongly on how their building blocks are connected. Simulating gas adsorption for every candidate framework is expensive, so a cheap structural descriptor that can rank frameworks before any simulation would be useful. In this work we build a pipeline that turns a MOF crystal structure into a coarse-grained graph of metal-oxo nodes and organic linkers, and then measures the heterogeneity of that graph with an influential-node multifractal analysis. The width of the resulting multifractal spectrum, $\Delta\alpha$, is used as the descriptor. We ran the pipeline on 150 structures returned by a live MOFX-DB query. The query asked for CoRE MOF 2019 entries with xenon isotherms, but the database returned hypothetical hMOF structures, which our provenance check flagged. Of these, 76 gave a usable spectrum. The values of $\Delta\alpha$ ranged from 1.032 to 1.582, with a narrow interquartile band and a tail of wide-spectrum frameworks. We also report three problems that limit the current results: $\Delta\alpha$ depends strongly on graph size, several structures are topological duplicates, and individual power-law fits are often poor. We have not yet tested whether $\Delta\alpha$ predicts any adsorption property.

***Index Terms*—Metal-organic frameworks, multifractal analysis, coarse-graining, network topology, MOFX-DB, hMOF.**

I. Introduction

Metal-organic frameworks (MOFs) are crystalline materials made from metal-containing nodes joined by organic linkers. Changing the node, the linker, or the way they connect gives a very large family of structures, and many of them are studied for gas storage and separation. The number of possible frameworks is far bigger than what anyone can simulate. Grand canonical Monte Carlo (GCMC) runs for adsorption are accurate but slow, and running them on every structure in a large database is not realistic.

A common way to deal with this is to filter first. If some inexpensive number computed from the structure alone could sort frameworks into groups, the expensive simulations could be limited to a smaller subset. Most simple descriptors of this kind, such as void fraction or pore diameters, come from the geometry of the empty space. In this project we look at a different side of the problem, which is the connectivity of the solid framework itself. We ask how uneven or heterogeneous the network of building blocks is, and whether a single multifractal number can capture that.

The idea is to represent each MOF as a graph. A single unit cell can have hundreds of atoms, and an atom-level graph mostly reflects bond chemistry inside each linker. Grouping atoms into chemically meaningful blocks, which is the coarse-graining step, gives a much smaller graph where each node is either a metal-oxo cluster or a whole organic linker. We then apply influential-node multifractal analysis (iNMFA) to this coarse-grained graph and use the width of the spectrum, $\Delta\alpha$, as a descriptor.

The contributions are as follows.

1) A reproducible pipeline from a live MOFX-DB query to a coarse-grained building-block graph, using a simple metal-oxo rule.
2) An application of iNMFA to the coarse-grained supercell graph, with repeated randomized box

coverings to reduce noise.
3) A data-driven grouping of 76 frameworks into Narrow, Medium and Wide classes by $\Delta\alpha$.
4) A list of the diagnostics that exposed the weaknesses of the method, including the finite-size effect, duplicate structures, and unstable fits.

II. Background and Related Work

A. Databases

CoRE MOF 2019 is a curated collection of experimental MOF crystal structures that were cleaned for computational use [3]. MOFX-DB collects structures together with simulated adsorption data and offers a web interface and a Python client [4]. The hMOF database is a large set of hypothetical MOFs generated computationally by combining building blocks [5]. Hypothetical structures are useful for screening studies, but they are not experimentally synthesized, and they are built from a limited library of blocks. This matters for our results, as discussed later.

B. Building-block extraction

MOFid gives a standard way to identify the metal nodes and organic linkers of a MOF [2]. MOFDiff uses coarse-grained building blocks as the representation for a generative diffusion model for MOF design [1]. In MOFDiff, each block is described by a set of connection points, and block identity is learned by a neural network. Our work borrows the general idea of splitting a framework into nodes and linkers, but we do not implement either MOFid or MOFDiff. Our partition rule is simpler and it is described in Section IV.

C. Fractal and multifractal analysis of networks

Box covering is a standard way to study self-similarity in networks. The network is covered with the smallest practical number of boxes of a given size, and the way the box count changes with box size is examined [7]. Multifractal analysis goes further by assigning a probability mass to each box and studying moments of that distribution at different exponents q [8]. If the mass distribution is uneven across the network, the resulting spectrum is wider. Influential-node multifractal analysis applies this idea only to a chosen subset of high-importance nodes, which reduces the cost and focuses the analysis on the hubs of the network. Our implementation follows the standard formalism, with the specific choices listed in Section IV.

TABLE I
Number of structures at each stage.

Stage	Count
Records returned by MOFX-DB	150
Connected periodic atom graph	77
Successful iNMFA spectrum	76
Failed (radius range too small)	1

III. Data

All structures were requested live from MOFX-DB through the `mofdb-client` Python package. The query asked for the database named CoRE MOF 2019 and required a xenon isotherm. We used the incremental `get_all` generator and stopped after 150 records, giving a candidate pool sorted by record identifier. No individual structure was selected by hand.

After download, our code checks the database field of each returned record against the requested one. This check showed that the server returned hMOF records and not CoRE MOF 2019 records. We do not know the reason for this mismatch. It may be a problem with how our query parameters are passed, a change in the server, or an issue in the client library. We have not yet resolved it. Because of this, all frameworks analyzed in this paper should be read as hypothetical hMOF structures.

Each record carries its CIF text. We write it to disk and read it with ASE [6]. A record is rejected if the CIF is empty, if the cell is not fully periodic, or if it has fewer than four atoms. Table I shows how the set shrank across the pipeline.

The drop from 150 to 77 is large. Those structures were removed at the parsing, connectivity, or supercell stages, and we did not keep a clean per-stage breakdown in the summary we are reporting from. The connectivity check is strict, and a disconnected periodic graph often points to an interpenetrated framework or leftover guest molecules in the CIF. We think this is the main reason, but we have not verified it structure by structure.

IV. Method

A. Periodic atom graph

Bonds are found with the ASE `neighbor_list` function under full periodic boundary conditions, using

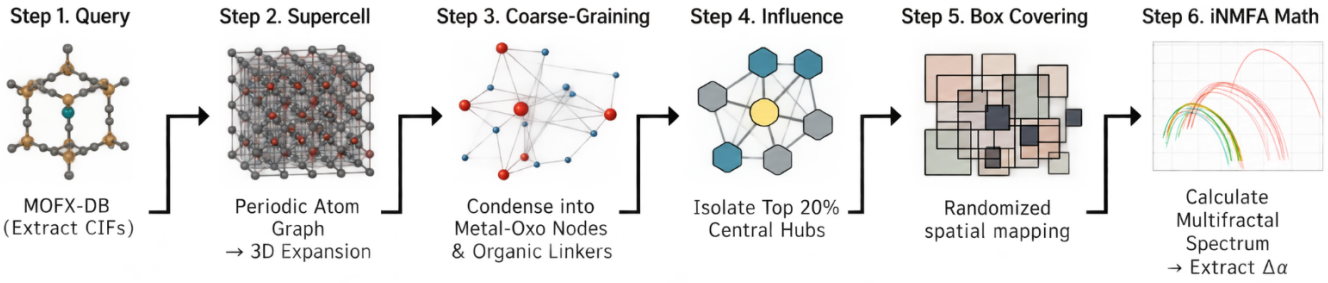


Fig. 1. Overview of the computational pipeline. Starting from a MOF structure retrieved from MOFX-DB, a periodic atomistic bond graph is constructed and expanded into a supercell. The structure is then coarse-grained into a building-block graph of metal-oxo nodes and organic linkers. The top 20 influential blocks are selected, and influential-node multifractal analysis (iNMFA) is performed via randomized box covering to obtain the multifractal spectrum and its width $\Delta\alpha$.

a cutoff that depends on the element pair. The cutoffs are 2.80 Å for metal-metal pairs, 2.45 Å for metal-nonmetal pairs, 1.70 Å for pairs made of C, N and O, 1.45 Å for any pair containing H or F, and 1.90 Å for everything else. The metal set is a fixed list of 33 elements common in MOF nodes. These cutoffs are practical choices and not fitted values. A structure whose periodic unit-cell graph is not connected is dropped.

Formally, let the unit cell have lattice matrix $L = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3]$ and atoms $i = 1, \ldots, n$ at Cartesian positions $\mathbf{x}_i$ with element Z_i. Atoms i and j are bonded if

$$\min_{\mathbf{t}\in\mathbb{Z}^3} \|\mathbf{x}_j + L\mathbf{t} - \mathbf{x}_i\| \le c(Z_i, Z_j), \tag{1}$$

where $\mathbf{t} = \mathbf{0}$ is excluded when $i = j$ and $c(\cdot,\cdot)$ is the pair cutoff listed above. This defines the adjacency matrix $A \in \{0,1\}^{n\times n}$, with $A_{ij} = 1$ for bonded pairs, and the degree $k_i = \sum_j A_{ij}$.

B. Supercell

Each unit cell is repeated until every lattice vector is at least 48 Å long, with a cap of 7000 atoms so that the all-pairs shortest-path computation stays affordable. If the cap would be exceeded, the repeat count is reduced one direction at a time, starting from the direction that contributes the most atoms. In symbols, the repeat counts are

$$n_k = \left\lceil \frac{48\ \text{Å}}{\|\mathbf{a}_k\|} \right\rceil, \qquad k = 1, 2, 3, \tag{2}$$

subject to $n\, n_1 n_2 n_3 \le 7000$, where n is the number of atoms in the unit cell. The supercell lattice is $L' = L\,\mathrm{diag}(n_1, n_2, n_3)$. The atom graph is rebuilt on the supercell with the same cutoffs and must again be connected. The supercell is built before coarse-graining. It gives the multifractal analysis a larger range of graph distances to work with.

C. Coarse-graining

The partition is done on the supercell atom graph. Every metal atom is put in a node set together with every oxygen atom directly bonded to it. The connected components of the subgraph induced by this set are the metal nodes. All remaining atoms are treated as organic, and the connected components of their induced subgraph are the linkers. Every atom therefore belongs to exactly one block. The coarse-grained graph G_{cg} has one vertex per block, and two blocks are joined if at least one atom-level bond crosses between them. No coordinates are computed for the blocks.

Written as sets, let M be the set of metal atoms and

$$S = M \cup \{\, v : Z_v = \mathrm{O},\ \exists\, m \in M \text{ with } A_{vm} = 1 \,\}. \tag{3}$$

The metal nodes are the connected components of $G[S]$ and the linkers are the connected components of $G[V \setminus S]$, where $G[X]$ denotes the subgraph induced by X. Together they form a partition $V = B_1 \cup \cdots \cup B_m$ with $B_a \cap B_b = \emptyset$ for $a \ne b$. Let $\pi : V \to \{1, \ldots, m\}$ map each atom to its block. The coarse-grained graph is $G_{cg} = (V_{cg}, E_{cg})$ with $V_{cg} = \{1, \ldots, m\}$ and

$$E_{cg} = \{\, \{\pi(u), \pi(v)\} : \{u, v\} \in E,\ \pi(u) \ne \pi(v) \,\}. \tag{4}$$

This rule is much simpler than MOFid. It only adds oxygen atoms one bond away from a metal, so more complex clusters can be split or mis-assigned. It also does no bond cutting and no embedding, which MOFDiff does. We therefore describe the step as a

simplified, MOFid-inspired partition and not as an implementation of either method.

A structure is kept only if G_{cg} is connected and satisfies $|V_{cg}| \geq 12$ and $|E_{cg}| \geq 8$.

D. Influential nodes

On the coarse-grained graph with N nodes we compute the all-pairs shortest path distance matrix D (unweighted). Nodes are ranked by degree, with closeness centrality breaking ties. The top 20 percent are taken as influential nodes. With D_{ij} the shortest-path length between blocks i and j, the closeness centrality is

$$C_i = \frac{N-1}{\sum_{j \neq i} D_{ij}}, \tag{5}$$

and nodes are sorted in decreasing lexicographic order of (k_i, C_i). The influential set $\mathcal{I}$ consists of the first $\lceil 0.2N \rceil$ nodes in this order. The graph diameter is $d = \max_{i,j} D_{ij}$. We used 20 percent and not the 10 percent we had used on atom graphs, because the coarse-grained graphs are much smaller.

E. Randomized box covering

For a box radius r, the graph is covered by boxes in the following way. Nodes are visited in a random order. If a visited node is not yet covered, it becomes a box center, and all still-uncovered nodes within distance r of it are put into that box. This continues until all nodes are covered. Because the visiting order is random, the covering changes from run to run. If c is a box centre and U is the set of nodes still uncovered when c is visited, the box is

$$\mathcal{B}(c, r) = \{\, v \in U : D_{cv} \leq r \,\}. \tag{6}$$

The boxes are disjoint and their union is V_{cg}. Let $N_B(r)$ be the number of boxes. For a fractal network the box count follows the power law [7]

$$N_B(r) \propto r^{-d_B}, \tag{7}$$

where d_B is the box-counting dimension.

The radii used are $r = 1, \ldots, r_{\max}$, where $r_{\max}$ is half of the graph diameter (rounded), with a minimum of 3 and a maximum of the diameter minus one. That is,

$$r_{\max} = \min(\max(\mathrm{round}(d/2),\, 3)\,,\; d-1)\,. \tag{8}$$

Structures with fewer than 3 usable radii are rejected, and this is what removed one framework in our run.

F. Multifractal quantities

For each box i that contains an influential node, we define the probability mass $p_i(r) = n_i(r)/N$, where $n_i(r)$ is the number of nodes in the box. The partition function is

$$P_q(r) = \sum_i [p_i(r)]^q\,, \tag{9}$$

where the sum runs over boxes containing influential nodes. We use 41 values of q evenly spaced from -10 to 10. For $q = 0$ the sum is defined as the number of such boxes, to avoid ambiguity with 0^0.

The mass exponent $\tau(q)$ is obtained from

$$\ln P_q(r) = \tau(q)\, \ln\!\Big(\frac{r}{d}\Big)\,, \tag{10}$$

where d is the graph diameter. We fit this as a straight line through the origin, so $\tau(q) = \sum xy / \sum x^2$ with $x = \ln(r/d)$ and $y = \ln P_q$. We use the through-origin form because in earlier tests a free intercept destroyed the peak of the spectrum at $q = 0$. The price is that the fit is forced, and we return to this in the discussion. The through-origin estimate is the least-squares solution

$$\hat{\tau}(q) = \arg\min_{\tau} \sum_r \left(y_r - \tau x_r\right)^2. \tag{11}$$

Setting the derivative with respect to τ to zero gives $\sum_r x_r\,(y_r - \tau x_r) = 0$, hence

$$\hat{\tau}(q) = \frac{\sum_r x_r\, y_r}{\sum_r x_r^2}. \tag{12}$$

The mass exponent is related to the generalized dimensions D_q by [8]

$$\tau(q) = (q-1)\, D_q, \tag{13}$$

so that $D_0 = -\tau(0)$ is the box-counting dimension of the boxes that contain influential nodes.

To reduce randomness, we run 8 independent batches, each averaging $\ln P_q(r)$ over 8 random coverings. The batch average

$$\overline{\ln P_q}^{(b)}(r) = \frac{1}{8} \sum_{m=1}^{8} \ln P_q^{(b,m)}(r), \qquad b = 1, \ldots, 8, \tag{14}$$

is used as y when $\tau(q)$ is fitted, once per batch. The singularity strength and the spectrum follow from a numerical derivative and a Legendre transform,

$$\alpha(q) = \frac{d\tau}{dq}, \qquad f(\alpha) = q\,\alpha(q) - \tau(q). \tag{15}$$

The spectrum width is $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$, computed for each batch and then averaged. The derivative is approximated by central differences on the grid $q_k = -10 + 0.5\,k$, $k = 0, \ldots, 40$,

$$\alpha(q_k) \approx \frac{\tau(q_{k+1}) - \tau(q_{k-1})}{2\,\Delta q}, \qquad \Delta q = 0.5, \quad (16)$$

with one-sided differences at the two ends, and the reported width is the batch mean

$$\overline{\Delta\alpha} = \frac{1}{B}\sum_{b=1}^{B} \Delta\alpha^{(b)}, \qquad B = 8. \quad (17)$$

We also record α_0, taken at the maximum of $f(\alpha)$, and an asymmetry

$$A = \ln \frac{\alpha_0 - \alpha_{\min}}{\alpha_{\max} - \alpha_0}, \quad (18)$$

which is negative when the spectrum is longer on the large-α side.

G. Properties of the spectrum

Three properties follow from the definitions above and are used when reading the results.

Peak at $q = 0$. Differentiating $f = q\alpha - \tau$ and using $\alpha = \tau'(q)$ gives

$$\frac{df}{d\alpha} = \frac{df/dq}{d\alpha/dq} = \frac{\alpha + q\,\alpha'(q) - \tau'(q)}{\alpha'(q)} = q. \quad (19)$$

The spectrum is therefore stationary exactly at $q = 0$, where $f(\alpha_0) = -\tau(0) = D_0$. Since $d^2 f/d\alpha^2 = 1/\alpha'(q) < 0$ whenever α decreases with q, this stationary point is a maximum.

Monotonicity. For each radius, $\ln P_q(r) = \ln \sum_i e^{q \ln p_i(r)}$ is a log-sum-exp in q and hence convex, and an average of convex functions is convex. Because $r < d$, every $x_r = \ln(r/d)$ is negative, so $\hat{\tau}(q) = \sum_r w_r\, y_r$ with $w_r = x_r / \sum x^2 < 0$ is a negative combination of convex functions and is concave. Hence $\alpha(q) = \hat{\tau}'(q)$ is non-increasing, and

$$\Delta\alpha = \alpha(-10) - \alpha(10). \quad (20)$$

Dependence on graph size. Every box contains at least one node, so $p_i(r) \geq 1/N$. For large negative q the partition function is dominated by the box of smallest mass, $P_q(r) \approx p_{\min}(r)^q$, which gives

$$\alpha(q) \approx \frac{\ln p_{\min}(r)}{\ln(r/d)} \leq \frac{\ln N}{\ln(d/r)}, \qquad q \to -\infty. \quad (21)$$

The left end of the spectrum can therefore move with $\ln N$ as the supercell grows. This is consistent with the finite-size effect reported in Section V-D, where enlarging the supercell raised $\Delta\alpha$ from about 0.54 to about 1.18.

Gaussian smoothing is applied only to the plots. All reported numbers come from the raw spectra.

H. Classification

Frameworks are sorted into three classes using the 25th and 75th percentiles of the $\Delta\alpha$ values in the set. Values at or below the lower percentile are Narrow, values above the upper percentile are Wide, and the rest are Medium. In symbols,

$$\text{class}(\Delta\alpha) = \begin{cases} \text{Narrow}, & \Delta\alpha \leq Q_{0.25}, \\ \text{Medium}, & Q_{0.25} < \Delta\alpha \leq Q_{0.75}, \\ \text{Wide}, & \Delta\alpha > Q_{0.75}, \end{cases} \quad (22)$$

where Q_p is the empirical p-quantile of $\Delta\alpha$ over the set. The thresholds come from the data, so they will move if the dataset changes.

I. Diagnostics

Besides the main run, the code performs a seed check, a finite-size check on one representative framework, and an isomorphism check that compares unit-cell atom graphs of equal size. We also looked at simple statistics of the coarse-grained graphs, namely clustering and degree assortativity.

V. Results

A. Spectra

All 76 spectra have their maximum of $f(\alpha)$ at $q = 0$, which is the expected shape for a well-behaved multifractal spectrum. The asymmetry A was negative for every framework. Fig. 2 shows the individual spectra on the left and the class averages on the right.

In the left panel, most Narrow and Medium spectra sit close together with peaks near α of 1.8 and $f(\alpha)$ between 2.1 and 2.4. The Wide spectra are shifted to larger α, with their maxima higher and their right ends reaching about 3.4. A few curves extend much further, and one reaches α close to 4. In the right panel the Narrow and Medium averages overlap over most of the range and cannot be separated cleanly. The Wide

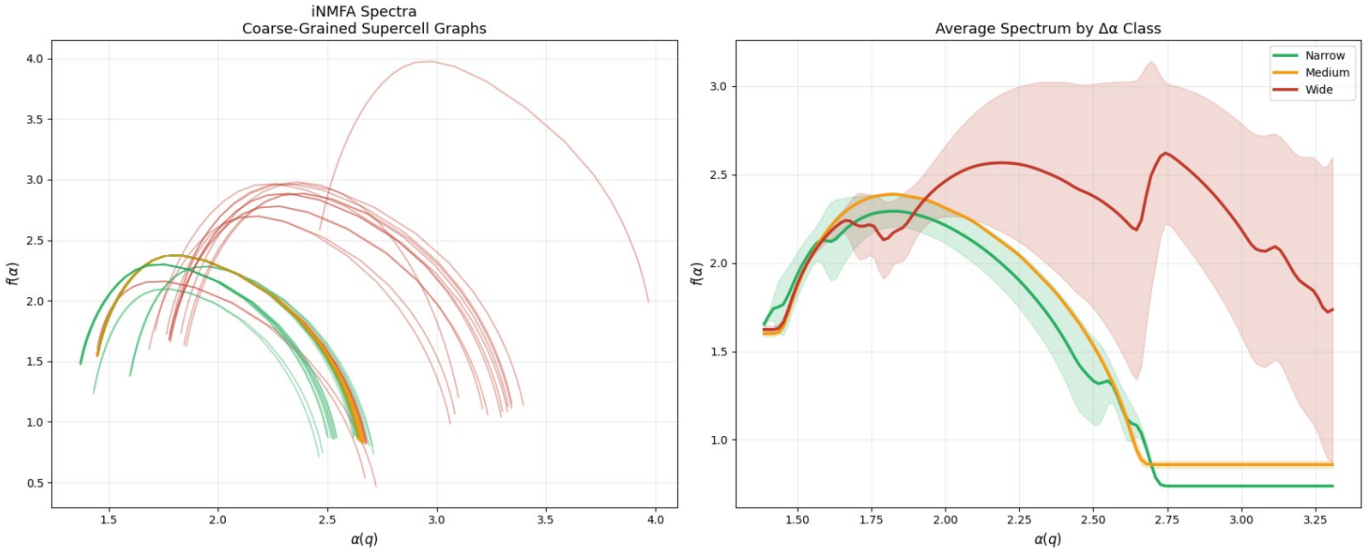


Fig. 2. iNMFA results on coarse-grained supercell graphs. Left: individual $f(\alpha)$ spectra, colored by class (green Narrow, orange Medium, red Wide). Right: class-average spectrum with one standard deviation band. Smoothing is for display only. The flat sections at the right end of the Narrow and Medium averages come from filling points outside each curve's range during averaging, so they should not be read as data.

TABLE II
CLASSES DEFINED BY PERCENTILES OF $\Delta\alpha$.

Class	Range of $\Delta\alpha$	Count
Narrow	≤ 1.190	19
Medium	1.190 to 1.222	38
Wide	> 1.222	19

average is clearly different, with a much larger spread. The bump near $\alpha = 2.75$ in the Wide average is probably caused by a small number of curves reaching that range, and we do not read anything into it.

B. Distribution of $\Delta\alpha$

Across the 76 frameworks, $\Delta\alpha$ ranges from 1.032 to 1.582. The middle of the distribution is tight, with the interquartile band running only from 1.190 to 1.222. Most frameworks are therefore very similar by this measure, and the Wide class is a tail above that core. Table II lists the classes.

The Narrow and Medium classes are separated by a very small difference in $\Delta\alpha$, since the lower edge of the middle band is 1.190 and the upper edge is 1.222. Percentile classes always give a 1:2:1 split by construction, so the class sizes carry no information. We would not claim that the boundary between Narrow and Medium is meaningful, and the overlap in Fig. 2 supports that caution.

C. Graph statistics

The coarse-grained graphs have clustering close to zero, meaning there are almost no triangles. This is expected for framework networks where blocks link to a small number of neighbors in an open lattice. Degree assortativity is strongly negative, so high-degree blocks tend to connect to low-degree ones. This fits the usual picture of metal nodes with many connections joined by linkers with two. It also means that much of the heterogeneity the method detects may come from the alternating node and linker structure that all frameworks share, and not from the differences between them.

D. Finite-size effect

This was the most important negative result. We repeated the analysis on the representative framework chosen by the code, using different supercell sizes. On the $1 \times 1 \times 1$ cell we obtained $\Delta\alpha \approx 0.54$, and on the $2 \times 2 \times 2$ cell the value was about 1.18. The framework is chemically the same in both cases, since the supercell only tiles the same blocks. So the value of $\Delta\alpha$ is not purely a property of the topology. It also depends on how many nodes are in the graph and on the range of radii that this allows. Any comparison of $\Delta\alpha$ between frameworks whose coarse-grained graphs have different sizes is therefore partly a comparison of graph size. Because our supercell rule fixes edge

lengths and not node counts, frameworks with small unit cells and frameworks with large ones end up with different numbers of blocks.

E. Duplicates

The isomorphism check found 5 groups of structures whose unit-cell atom graphs are isomorphic, containing 17 structures in total. These are topological duplicates, at least at the level of the atom graph. The effective number of distinct frameworks is therefore lower than 76. We did not remove them before computing the class thresholds, so the percentiles are also affected. Note that hypothetical databases are built from a limited set of blocks and nets, which makes such repeats more likely.

F. Quality of the fits

The through-origin R^2 for individual q values is often poor. The mean R^2 of a framework was negative in many cases, with one example of -1.17. Negative values are possible because we compute R^2 against the mean of the data while forcing the line through the origin, so a fit can be worse than a horizontal line. Explicitly,

$$R^2 = 1 - \frac{\sum_r (y_r - \hat{\tau} x_r)^2}{\sum_r (y_r - \bar{y})^2}, \tag{23}$$

and because the model has no intercept the numerator can exceed the denominator, which gives $R^2 < 0$. It does not necessarily mean that the box-counting data has no scaling. It does mean that individual slices of $\tau(q)$ are noisy, even though the averaged spectra look smooth and well shaped. The smooth look of the spectra should therefore not be taken as proof that each power-law fit is good. The number of radii used is small, especially for the smaller graphs, and this contributes to the problem.

VI. Discussion

The pipeline works mechanically. It goes from a database query to a coarse-grained graph to a well-shaped multifractal spectrum for 76 structures without manual selection. However, the results do not yet support using $\Delta\alpha$ as a reliable descriptor, for the following reasons.

First, the data are not what we intended. The results describe hypothetical hMOF structures, so we cannot say anything about experimental frameworks. Fixing the query, or switching to another endpoint, is the first thing to do.

Second, the finite-size dependence is severe. A change of about 0.6 in $\Delta\alpha$ between two supercell sizes of one framework is larger than the whole range we observed across the set, which is about 0.55. This means that the differences between frameworks could be dominated by the graph size. A fair comparison needs either graphs of matched size, or a fit window that is normalized in a way that removes this dependence. We have not done either yet.

Third, many frameworks look alike. The middle 50 percent of $\Delta\alpha$ values lies in a band of width about 0.03, and 17 structures are duplicates. The descriptor may separate only a small number of unusual frameworks from a large uniform group.

Fourth, we discard geometry after the graph is built. Bond lengths, angles, pore shapes, and the chemistry of the linkers do not enter the graph. Two frameworks with the same connectivity but very different pores would receive the same score. Adsorption depends strongly on pore geometry, so this is a real limitation for the intended use.

Fifth, the coarse-graining rule is simplified. The single-hop metal-plus-oxygen rule can mis-assign atoms in complex clusters, which changes the coarse-grained graph and therefore the spectrum.

Finally, we have not yet checked whether $\Delta\alpha$ correlates with anything measurable. The database provides xenon isotherms, and the void fraction, the largest cavity diameter and the pore limiting diameter are available for the records, so this test is possible. Without it, we cannot say that the descriptor is useful for filtering. The original motivation of the project remains untested.

VII. Future Work

The next steps, in the order we plan to do them, are the following.

1) Fix the query so that real CoRE MOF 2019 structures are returned, and confirm this with the provenance check.
2) Remove the size confound, either by comparing graphs with similar numbers of nodes or by changing the fit window, and repeat the finite-size test on several frameworks and not only one.

3) Remove duplicate structures before defining the percentile classes.
4) Test whether $\Delta\alpha$ correlates with pore metrics and with simulated xenon uptake.
5) Improve the fit quality by using more radii, and examine whether a free intercept can be used without losing the peak at $q = 0$.
6) Consider a better partition rule closer to MOFid, and add simple geometric or chemical descriptors alongside the topological one.

VIII. Conclusion

We built a pipeline that takes MOF structures from a live database query, converts them into coarse-grained building-block graphs on supercells, and measures the multifractal spectrum width of the influential nodes. On 76 hypothetical hMOF structures the method gave well-shaped spectra that all peak at $q = 0$, with $\Delta\alpha$ between 1.032 and 1.582 and most values in a narrow band. At the same time, the work showed clear limits. The database did not return the experimental set that we requested, $\Delta\alpha$ changed from about 0.54 to about 1.18 when the supercell of one framework was enlarged, 17 structures were topological duplicates, and individual fits were often poor. For these reasons we regard $\Delta\alpha$ as a descriptor that is not yet validated. The pipeline is a usable base, and the next tasks are to obtain the correct data, remove the dependence on size, and test the descriptor against adsorption data.

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